Closed-loop learning clustering and data autophagy

Gabriele Cerioli
Supervisor: Matteo Marsili

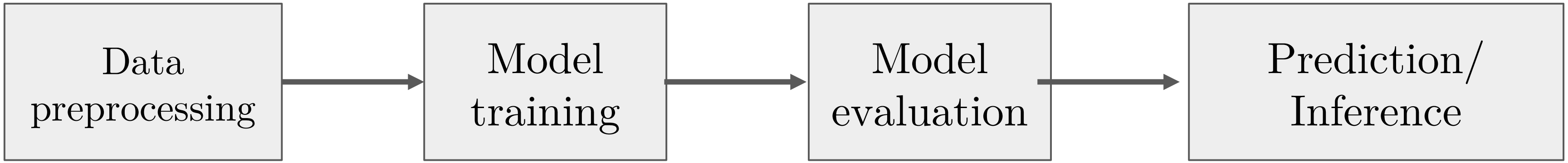
ICTP
The Abdus Salam
International Centre
for Theoretical Physics

UNIVERSITAS STVDIORVM
TERGESTVM
MCMXXIV

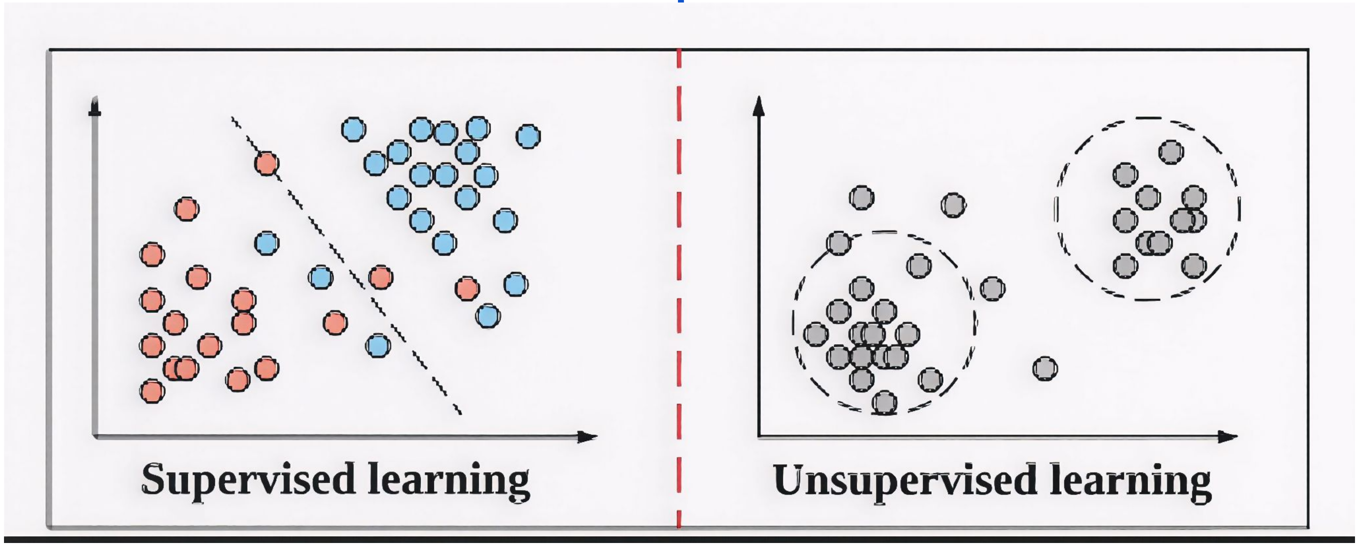
Contents

What is machine learning?

- Algorithms that allow computer to **learn from data**
- Improve performance on a task without being explicitly programmed

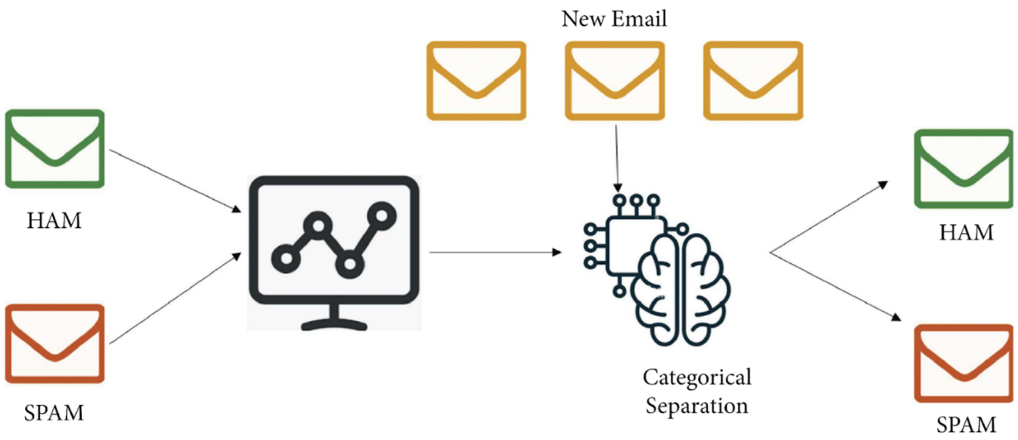


Supervised learning	Unsupervised learning
The model is trained on **labeled data**	Must find **hidden patterns** in the dataset **on its own**

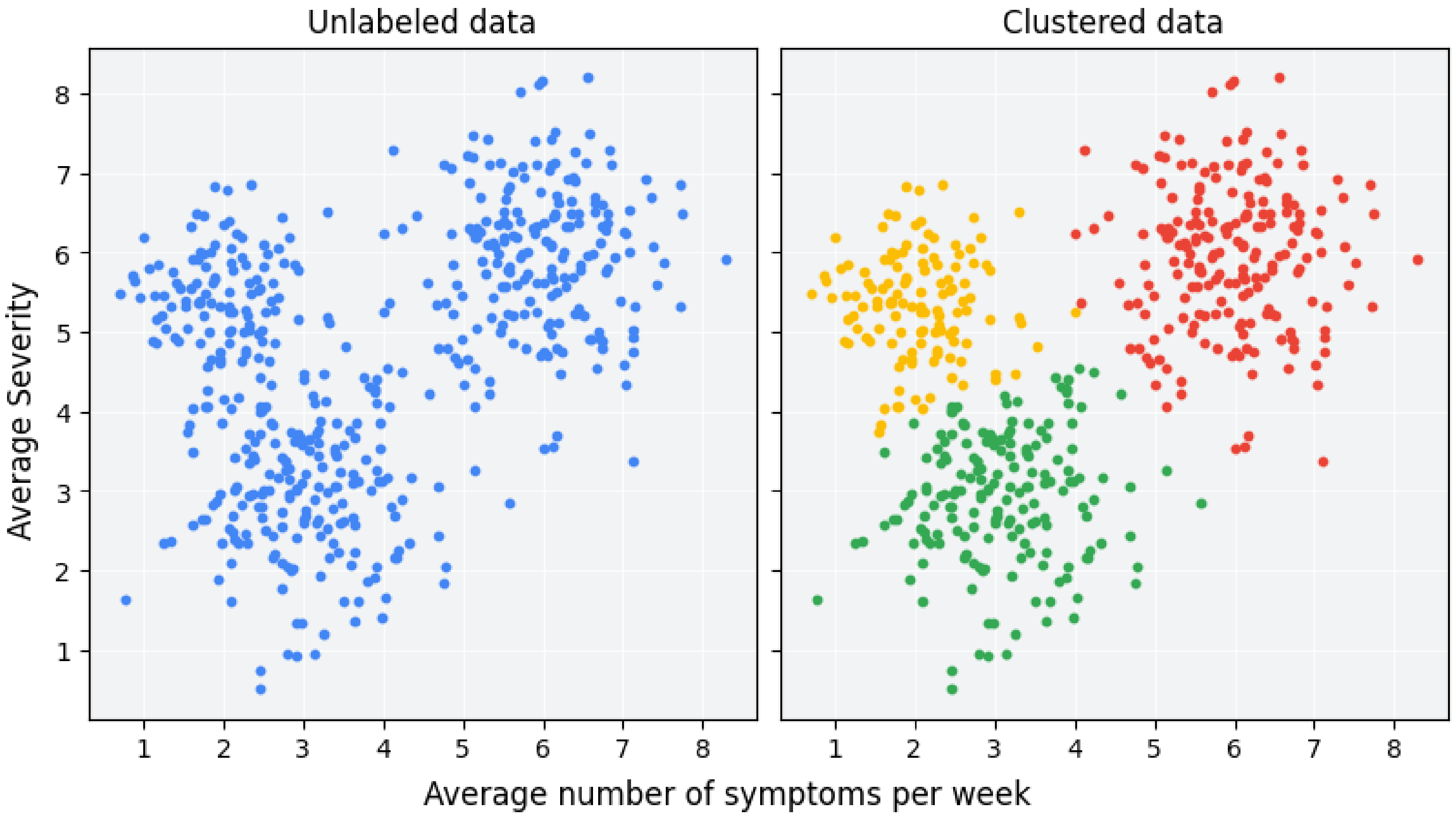


Supervised learning: Email spam classification

Data preprocessing → Train with million of spam and non spam emails → Test it with new emails → We can now use the model

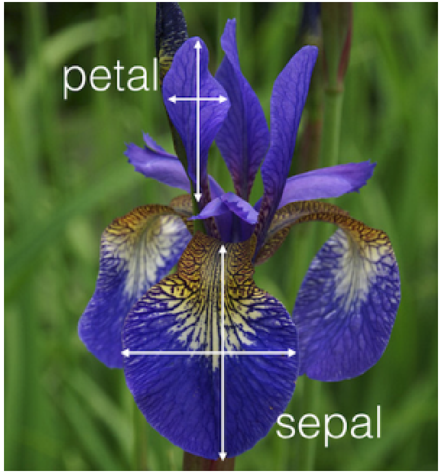


Unsupervised learning: Clustering problems

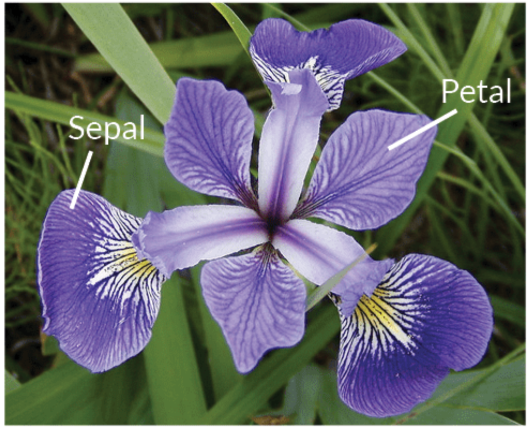


Iris flower dataset: an interesting playground

- Data about three different plant species collected in 1936 by Ronald A. Fisher
- 150 samples with 50 samples for each species of Iris flowers



We have **4 measures** for each sample



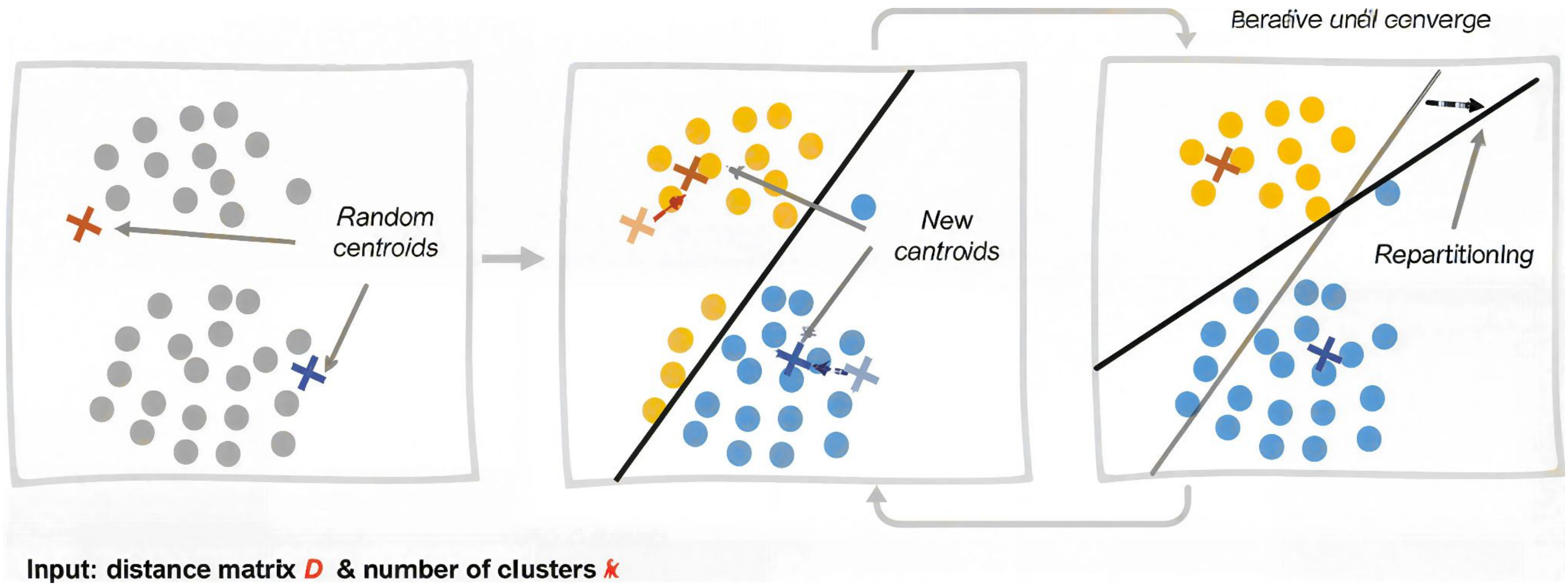
Iris Versicolor

Iris Setosa

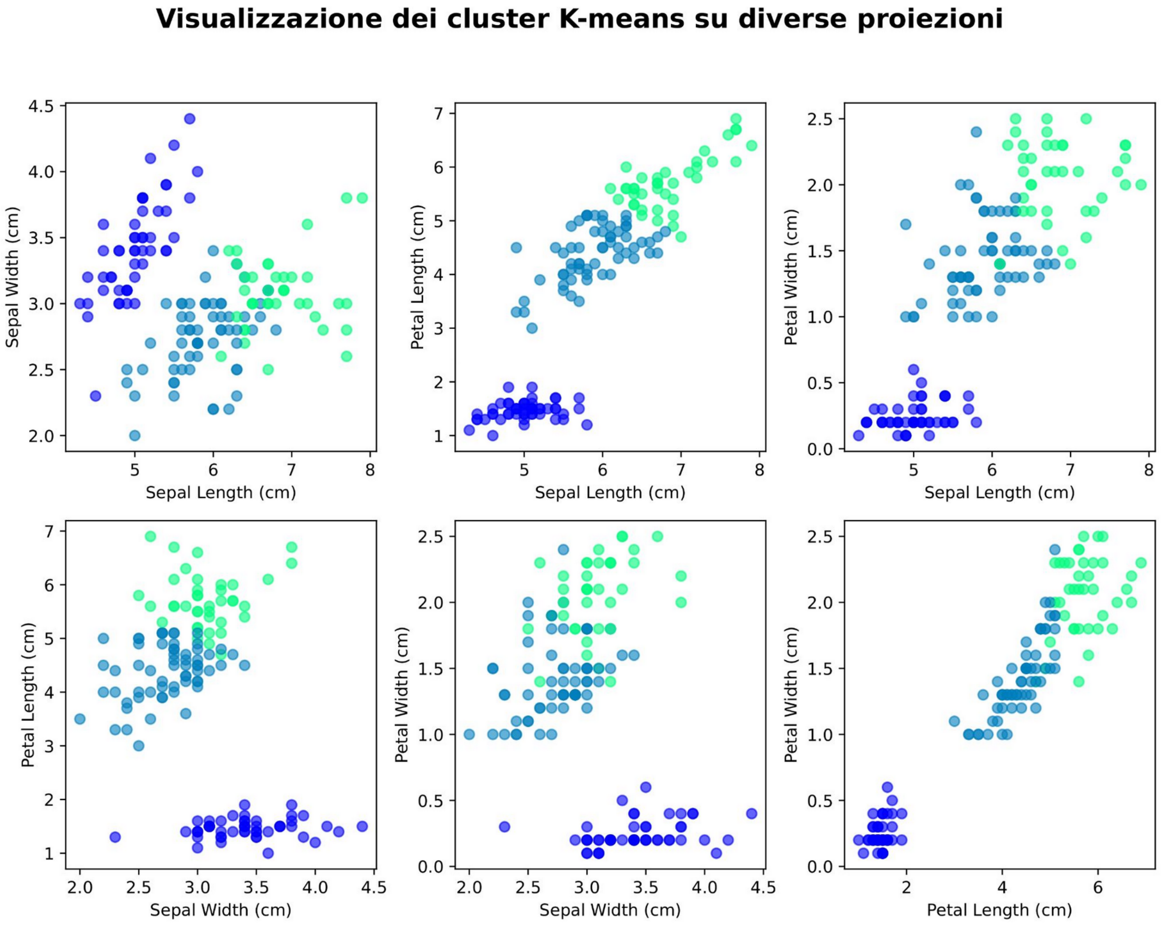
Iris Virginica

K-means algorithm

1) Initialization: choose K random centroids
2) Assignment step: assign each point to the closest cluster
3) Update step: centroids are recalculated using the mean of each cluster
4) REPEAT TO CONVERGENCE



K-means on Iris dataset



Soft/Fuzzy K-means algorithm

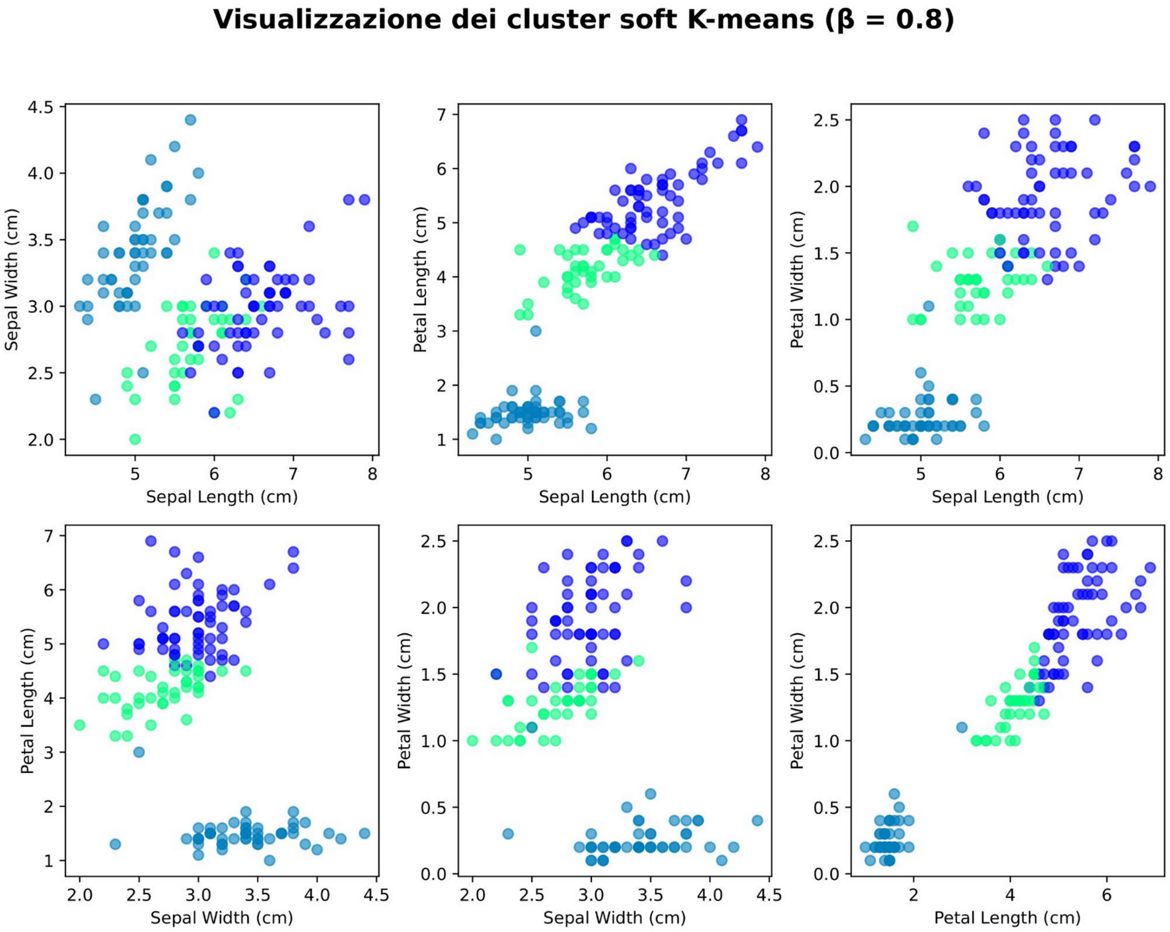
The main difference is that we don't force each data point to belong to only one cluster (hard assignments) but we define a **degree of membership (responsibility r)** to each cluster.

$$r_k^{(n)} = \frac{e^{-\beta\, d(m^{(k)}, x^{(n)})}}{\sum_{k'} e^{-\beta\, d(m^{(k')}, x^{(n)})}}$$

$$m^{(k)} = \frac{\sum_n r_k^{(n)} x^{(n)}}{\sum_n r_k^{(n)}}$$

Now the updating of the centroids is a **weighted mean**, where the weights are the responsibilities r.

Soft K-means on Iris Dataset with stiffness $\beta = 0.8$



$\beta = 0.8$

How to evaluate the clustering quality?

Purity Score

To calculate the purity score we need the ground truth (c_j)

$$P = \frac{1}{N}\sum_{k=1}^{K}\max_j |\omega_k \cap c_j|$$

ω_k is the set of points in the cluster k

Davies-Bouldin Index (internal metric)

We first need to define the qth moment of the points in cluster i about the mean:

$$S_i = \left(\frac{1}{T_i} \sum_{j=1}^{T_i} |X_j - A_i|^q \right)^{\frac{1}{q}}$$

where A_i is the centroid of cluster i and T_i is the number of points in cluster i.

Clusters i and j are characterized by the Minkowski metric:

$$M_{ij} = \left(\sum_{k=1}^{N} |A_{ik} - A_{jk}|^p \right)^{\frac{1}{p}}$$

(for $p = 1$ M_{ij} is the Mahattan distance, for $p = 2$ M_{ij} is the Euclidian distance)

To define a metric R_{ij} we ask these properties:

- $R_{ij} \geq 0$
- $R_{ij} = R_{ji}$
- When $S_j \geq S_k$ and $M_{ij} = M_{ik}$ then $R_{ij} > R_{ik}$
- When $S_j = S_k$ and $M_{ij} \leq M_{ik}$ then $R_{ij} > R_{ik}$

We can now define the similarity between clusters:

$$R_{ij} = \frac{S_i + S_j}{M_{ij}} \tag{5}$$

Of course it is ideal that M_{ij} (separation between two clusters) is as large as possible, and S_i as low as possible.

For each cluster we calculate the maximum of this value:

$$R_i = \max j \neq i R_{ij}$$

We can now define the Davies-Bouldin index for N clusters:

$$\overline{R} = \frac{1}{N} \sum_{i=1}^{N} R_i$$

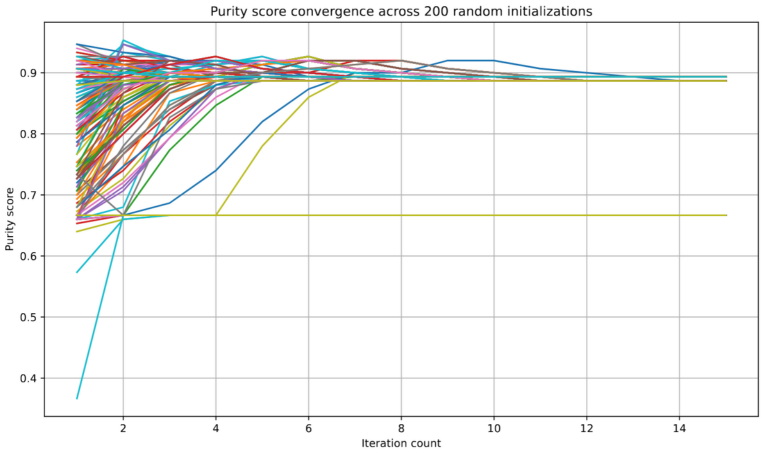
We expect the index to be as little as possible

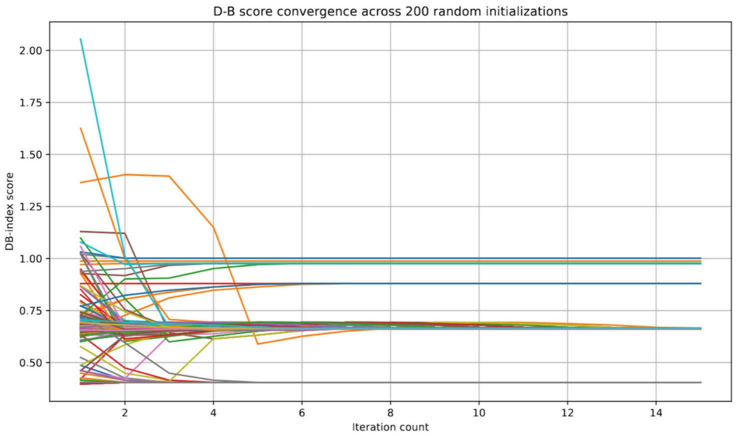
Infomax based metric $\hat{H}[s]$

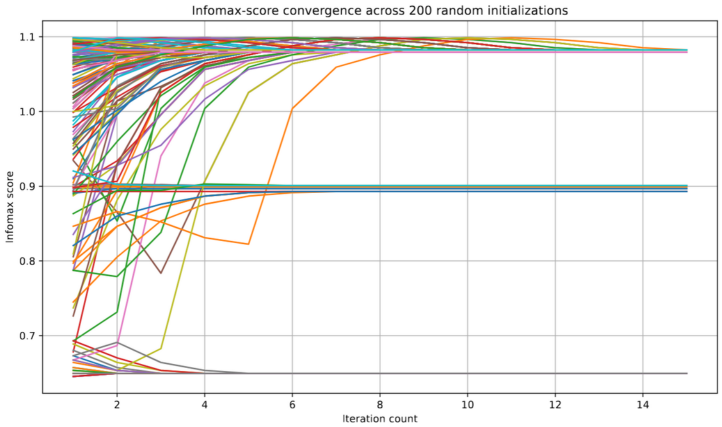
Since we know that the information collected by the dataset can be quantifies by the Shannon entropy, we can take the entropy as a metric:

$$\hat{H}[s] = -\sum_{s} \frac{K_s}{N} \log\left(\frac{K_s}{N}\right)$$

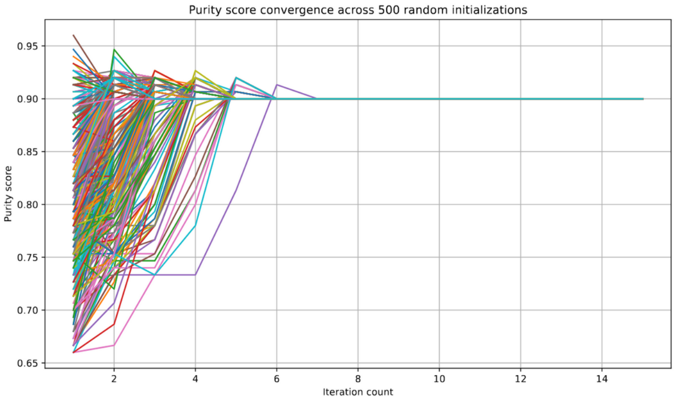
Kmeans evaluation

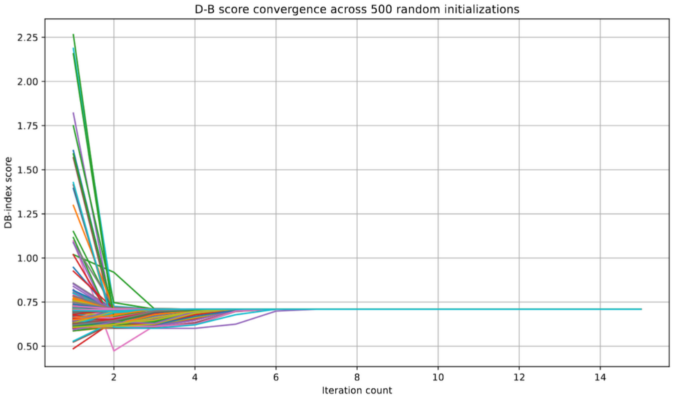


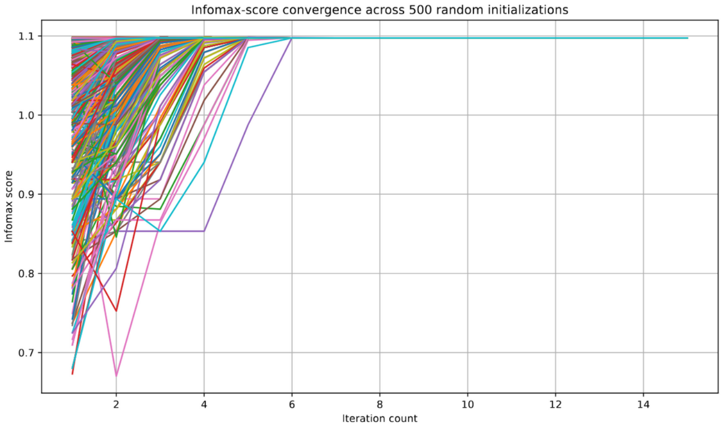




Soft Kmeans evaluation







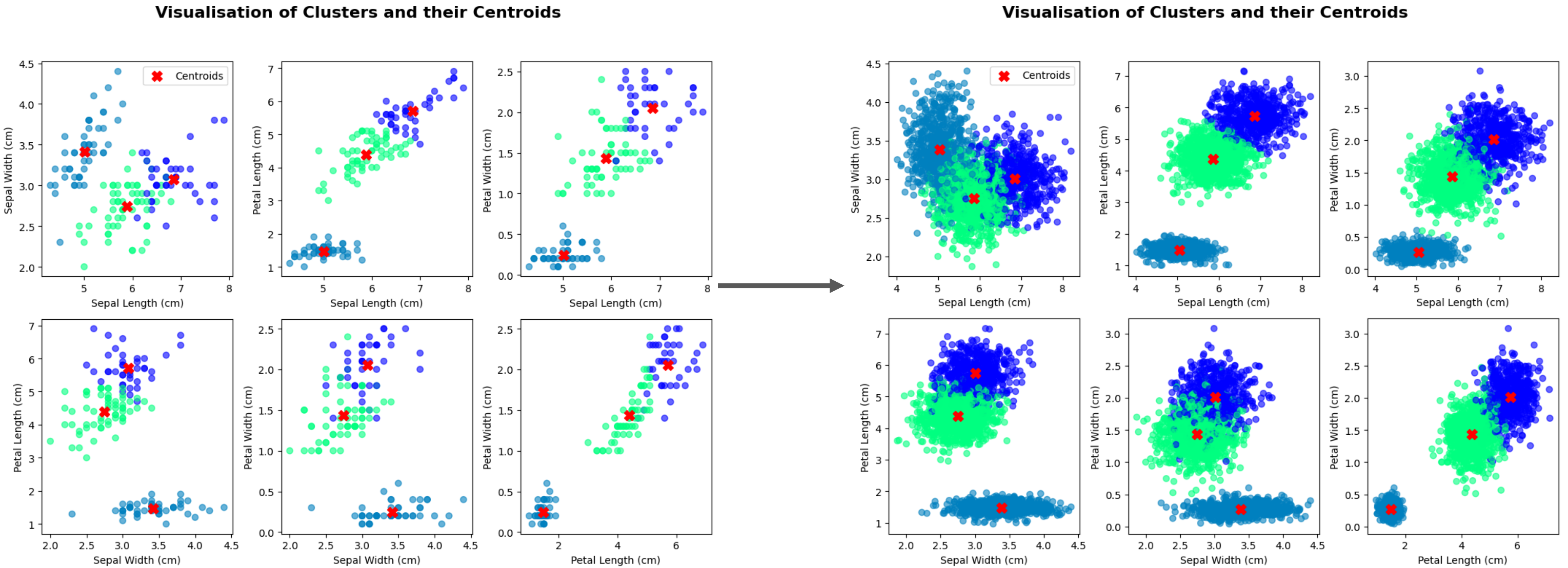
What is Closed-loop learning?

Open-Loop

Closed-Loop

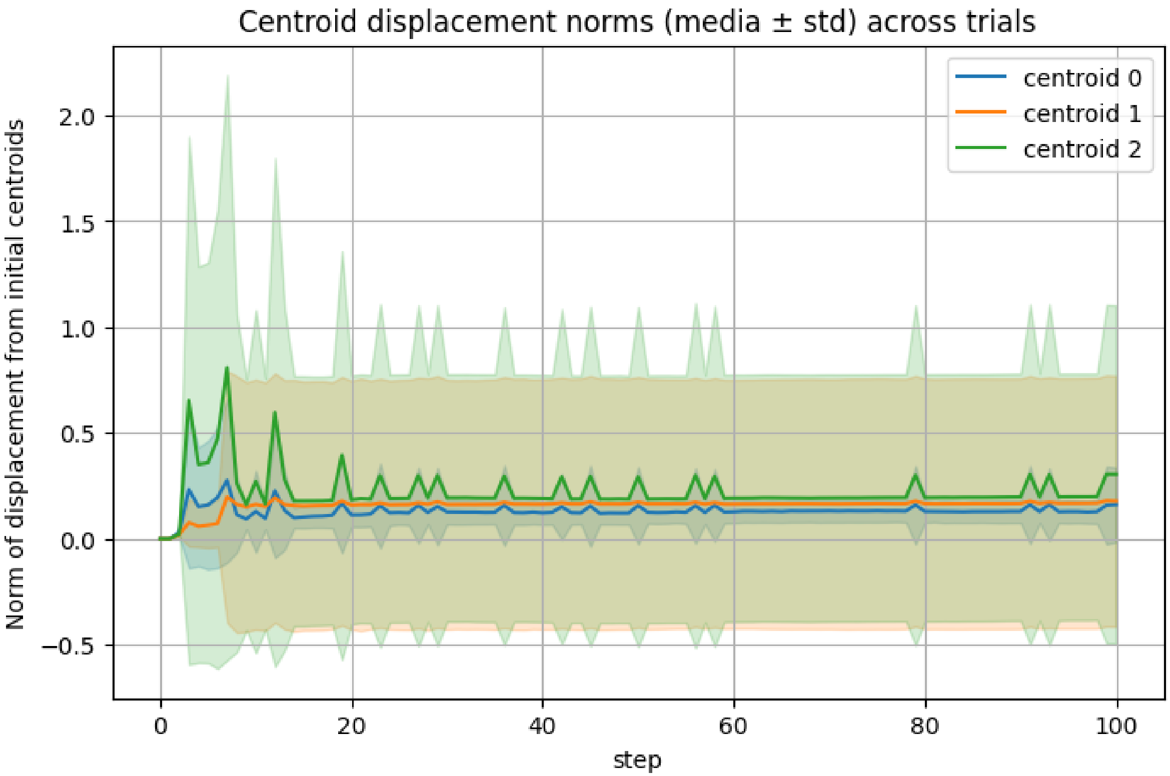
Our Closed-loop learning algorithm

After doing a first K Means clustering, we **add M synthetic points** (generated from the clusters Gaussian mixture), and repeat this cycle multiple times.



CLL as an evaluation of clustering quality

Firstly we compute the displacement of the centroids after the CLL execution.



CLL as an evaluation of clustering quality

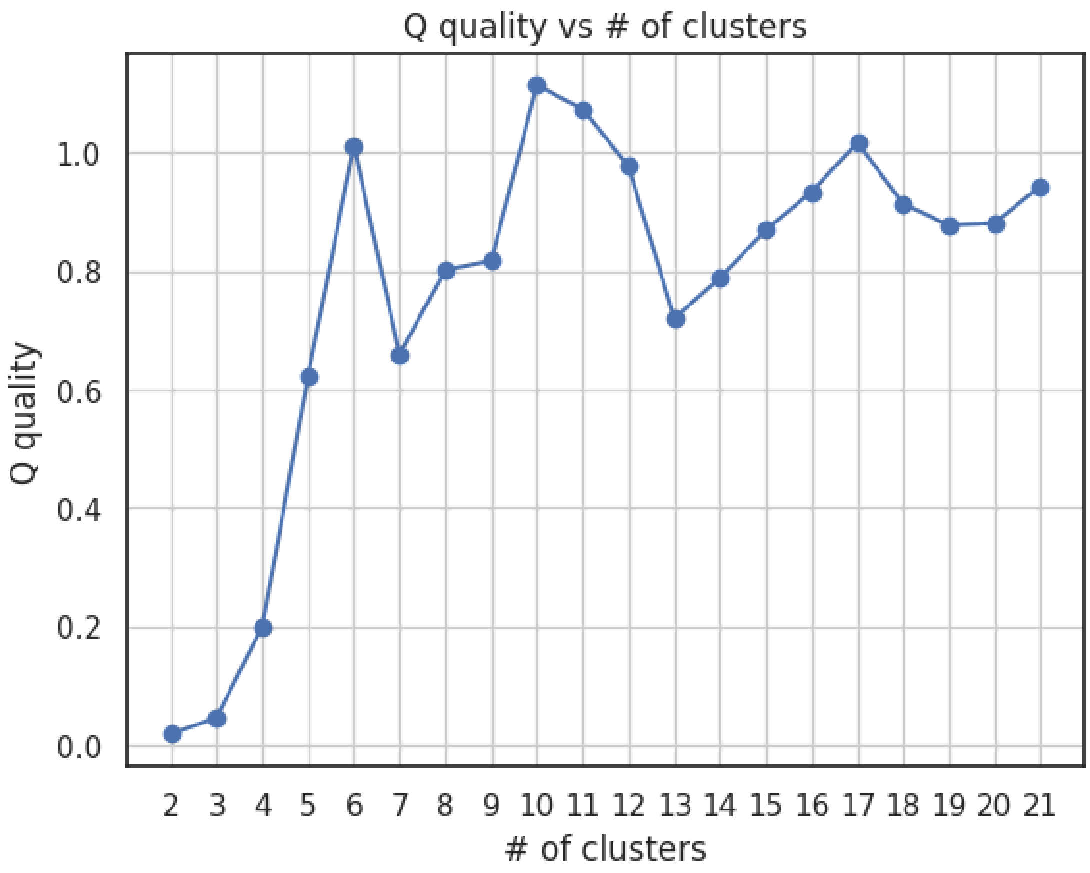
Then we confront the displacements with the distances between the original centroids

Given N clusters we can define the cluster quality Q as:

$$Q = \frac{\sum_{i=1}^{N} \frac{\delta_i}{D_i}}{N}$$

We expect this value to be as low as possible for a good clustering

Can we use the CLL quality to find out the optimal number of clusters?



CLL as an inference device

The problem is the following:

- Create a sample of N points using the normal distribution with randomly generated parameters
- Make a Likelihood estimation of the parameters
- Create M synthetic points from a normal distribution with the parameters estimated above
- Repeat this cycle with progressively larger dataset

CLL as an inference device

If we define:

- μ as the original random generated mean
- $\hat{\mu}$ as the mean estimated using simple Maximum Likelihood
- $\tilde{\mu}$ as the mean estimated after CLL cycle

and the same thing for the standard deviations.
We can define:

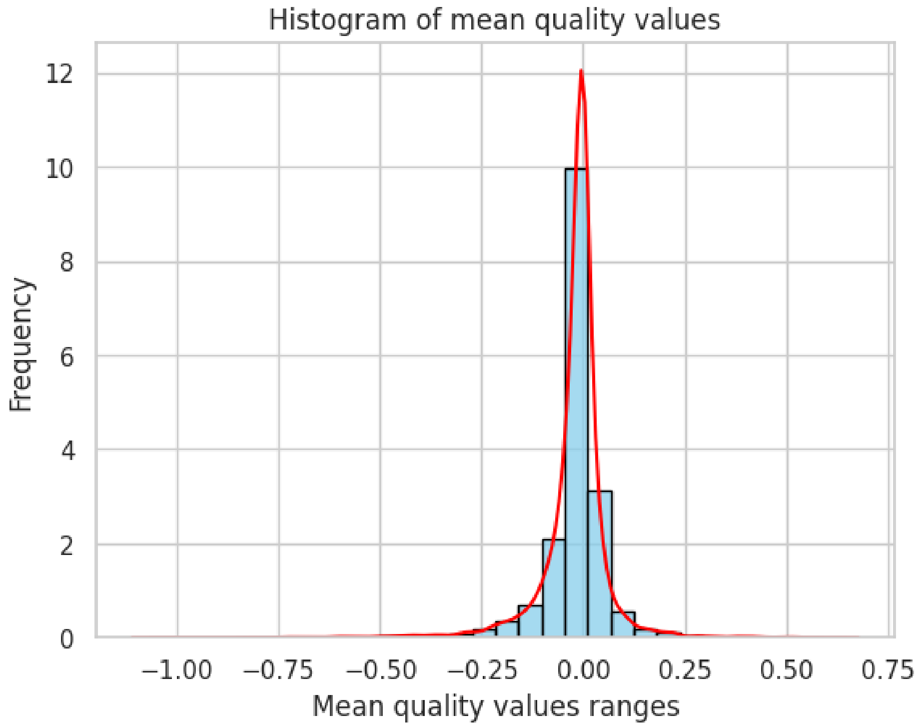
$$\chi_\mu = \frac{|\hat{\mu} - \mu| - |\tilde{\mu} - \mu|}{\mu}$$

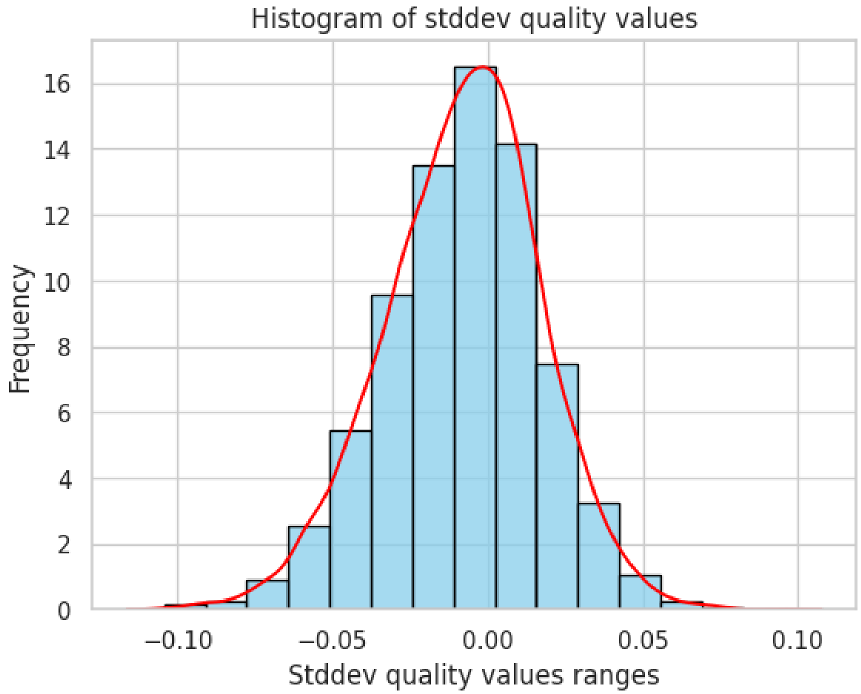
$$\chi_\sigma = \frac{|\hat{\sigma} - \sigma| - |\tilde{\sigma} - \sigma|}{\sigma}$$

Of course if χ is negative, CLL has made our estimation of the parameter worse.

CLL as an inference device

If we iterate this algorithm 1000 times we get:



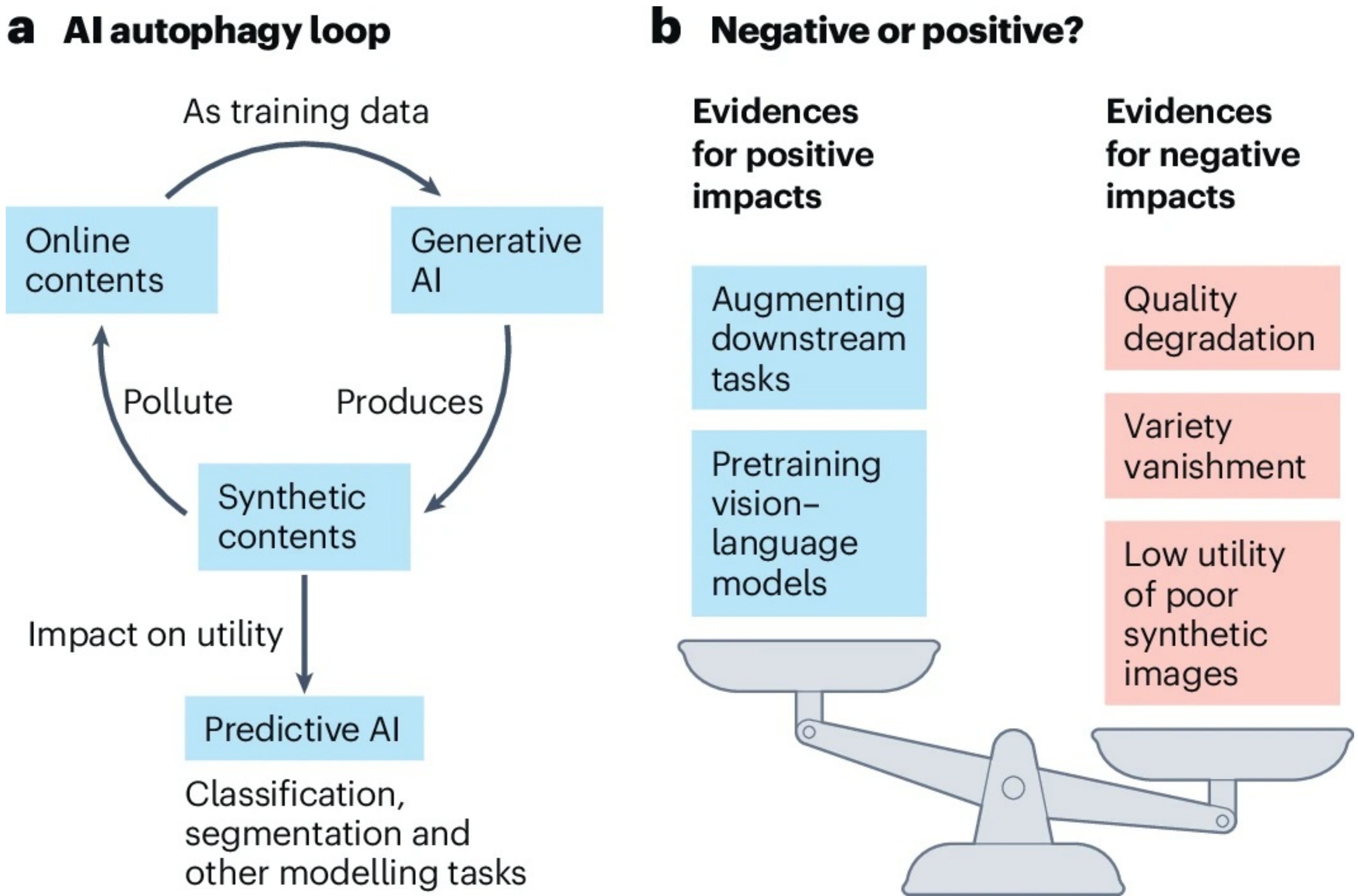


So it seems it doesn't work...

Data Autophagy

What is Data Autophagy?

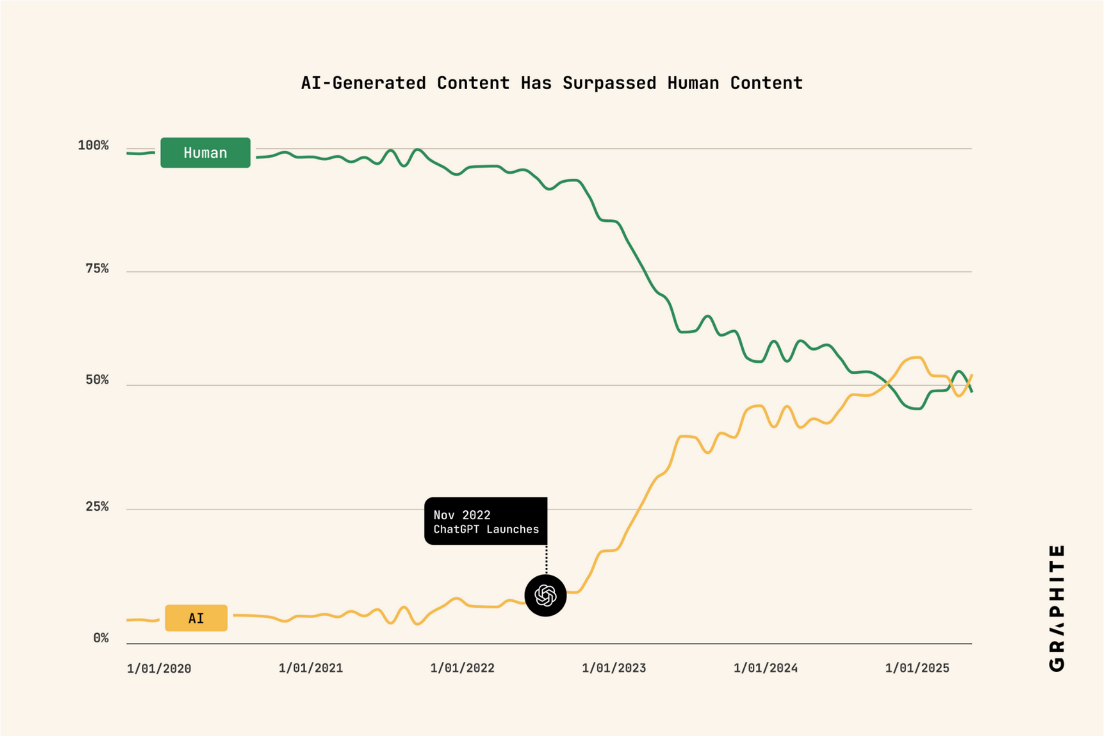
When generative AI models are increasingly trained on content that was also AI generated



Why is it a problem?

The supply of high-quality human generated data is running out.

A possible outcome of data autophagy is **model collapse**, where the models lose the diversity of the original data.



How can we prevent the inevitable model collapse?

GOOD NEWS

- If you train a model using **Maximum Likelihood** on purely self-generated Data the model collapse is inevitable
- If you use a **Maximum A Posteriori (MAP) regularization**, it introduces a small external force that prevents model collapse
- If you include a **small amount of non-AI-generated data** (even one point) it is enough to prevent model collapse

BAD NEWS

$$f(s|\theta) = \frac{1}{Z(\theta)} e^{\sum_a \theta_a \phi_a(s)}$$

Where $\phi_a(s)$ are bounded functions of s

- It is demonstrated to only work for **Exponential Family models** by Marsili-Roudi-Jangjoo (July 2025)